

Technical Design Document — Layout & UI Design

Restaurant Management System (RMS) — Phase 1 Desktop (JavaFX)

Document	Layout & UI Design TDD
Version	1.0 (Draft for review)
Date	11 July 2026
Classification	Confidential
Prepared by	Project Team
Traces to	BRD v1.0 · FRD v1.0 · TDD v1.0
Platform	JDK 1.8 (Java 8) · JavaFX 8
Phase	Phase 1 (Desktop) → Phase 2 (Web)

1. Introduction

1.1 Purpose

This document is the **presentation-layer (View) companion** to the RMS Technical Design Document (TDD v1.0). Where the TDD defines the layered architecture, database schema, services and algorithms, this document specifies **how the JavaFX desktop user interface is laid out**: the screen architecture, the navigation shell, the reusable layout patterns, the per-screen wireframes for every module, and the concrete JavaFX frameworks, libraries and tooling used to build them.

It is written so that a developer can open Scene Builder, create the FXML files described here, wire them to the controllers named in TDD §2.3, and produce screens that satisfy the functional requirements (FR-01...FR-30) and the usability requirement (NFR-01).

1.2 Scope

Covered here: the JavaFX **View and Controller** tiers only (TDD §2.2). The Service, DAO and Database tiers are unchanged and are not restated. Every screen maps to functional

requirements already specified in the FRD, and every layout decision is constrained by three things carried over from the parent documents:

- **NFR-01 Usability** — front-line POS screens must be learnable in one short session, common actions in minimal clicks.
- **NFR-05 / NFR-06 Migration readiness** — the View must stay "thin": no business logic, no SQL, no calculation in controllers, so Phase 2 can swap JavaFX for a web front end without touching the reusable core.
- **RBAC (FR-02)** — the interface shows/hides/disables functions by the logged-in role (Administrator, Manager, Cashier).

1.3 Relationship to other documents

Ref	Document	What this doc takes from it
REF-1	BRD v1.0	Scope, roles, business objectives.
REF-2	FRD v1.0	Screen behaviour, field validation (Appendix A), role-permission matrix (§2.4), state models, enumerations (Appendix B).
REF-3	TDD v1.0	Layered architecture, <code>com.rms</code> package structure, service/controller names, data dictionary.
REF-4	This document	Screen layout, navigation, JavaFX libraries and UI patterns.

1.4 Definitions

Term	Meaning
FXML	XML markup that declares a JavaFX scene graph; the "View" file, editable in Scene Builder.
Controller	The Java class bound to one FXML file via <code>fx:controller</code> ; handles UI events only.
Scene graph	The tree of JavaFX nodes (panes, controls) that make up a screen.
Layout pane / container	A node that arranges its children (<code>BorderPane</code> , <code>GridPane</code> , <code>VBox</code> , <code>HBox</code> , <code>StackPane</code> , etc.).
View manager	A small helper that loads FXML and swaps the center content of the shell.
Shell	The persistent application frame (top bar + side navigation + content area) that hosts every module screen.
CSS	JavaFX cascading style sheet (<code>.css</code>) controlling colours, fonts, spacing — the single source of visual styling.

1.5 Revision history

Version	Date	Description
0.1	10 Jul 2026	Outline: shell, navigation, library shortlist.
1.0	11 Jul 2026	Full layout design with per-screen wireframes and verified Java 8 library set.

2. UI Design Goals & Principles

The layout follows six principles, each traceable to a parent requirement:

1. **Thin View, no logic (NFR-05/06).** FXML declares structure; controllers only capture input, call a service, and render the result. No arithmetic, no SQL, no state rules in the UI. This is what lets Phase 2 reuse the whole core.
2. **Speed for the front line (NFR-01, BO-1).** The POS order screen is optimised for the most frequent path — pick table → tap items → take payment — with large touch-friendly buttons and a permanently visible running total.
3. **One consistent shell.** Every module opens inside the same frame (top bar, left navigation, content area). Users learn the frame once; only the center changes.
4. **Role-aware surface (FR-02).** Navigation items and action buttons a role cannot use are hidden or disabled — defence in depth pairs with the business-layer RBAC check.
5. **Status made visible (FR-09, FR-22).** Table status, order status, low-stock and reservation status are shown with consistent colour coding, never colour alone (icon + text label for accessibility).
6. **Fail-safe forms.** Inline field validation (FRD Appendix A) with clear messages; destructive actions confirm; nothing is lost silently. Matches the FRD's standardised error messages.

3. Presentation-Layer Technology Stack (JavaFX 8 / JDK 1.8)

3.1 Core platform

Concern	Choice	Notes
Runtime	JDK 1.8 (Java 8, ≥ 8u60)	JavaFX 8 ships <i>inside</i> the Oracle/Zulu FX JRE — no separate JavaFX dependency needed on Java 8. (JFoenix requires 8u60+.)
UI toolkit	JavaFX 8 (<code>javafx.*</code> , bundled)	Scene graph, controls, properties/bindings, <code>Task</code> / <code>Service</code> for background work.
Markup	FXML	One <code>.fxml</code> per screen; keeps structure out of Java, editable visually.
Visual design tool	Gluon Scene Builder 8.5.0	The last Scene Builder line that targets JavaFX 8. Drag-and-drop FXML editing.
Styling	JavaFX CSS	A single <code>rms.css</code> theme + a few component classes. No inline styling.
Build	Maven (or Gradle)	Dependency management for the libraries below.

Important Java 8 constraint. Several popular JavaFX libraries dropped Java 8 support and now require Java 11+. On JDK 1.8 you must pin the **last 8-compatible version** of each. The table in §3.2 lists those exact versions; §3.3 lists what to avoid.

3.2 Recommended external libraries (all verified Java 8 / JavaFX 8 compatible)

Purpose	Library	Java 8 version	Maven coordinates	Why / where used
Material-style controls (buttons, text fields with floating labels, dialogs, spinners)	JFoenix	8.0.10	<code>com.jfoenix:jfoenix:8.0.10</code>	Modern look for login, f buttons. The 9.x line is so pin 8.0.10.
Advanced controls (notifications, <code>NotificationPane</code> , <code>StatusBar</code> , <code>MaskerPane</code> , <code>SegmentedButton</code> , <code>CheckComboBox</code> , validation decorations)	ControlsFX	8.40.18	<code>org.controlsfx:controlsfx:8.40.18</code>	Toasts, busy overlays, f tables, decoration-base validation. Last 8-series
Icons	FontAwesomeFX	8.15	<code>de.jensd:fontawesomefx-fontawesome:4.7.0-9.1.2</code> (or the <code>de.jensd:fontawesomefx:8.15</code> bundle)	Scalable vector icons f and buttons. Classic Java choice. <i>Alternative: Ikonli</i> (<code>org.kordamp.ikonli:javaafx:1.9.0</code> + a pack Ikonli 12.x is Java 11+, 1.9.0 if you prefer it.
PDF receipts & report export (FR-16, FR-30)	JasperReports	6.20.x	<code>net.sf.jasperreports:jasperreports:6.20.6</code>	Designed receipt/report templates → PDF/print. <i>alternative: OpenPDF</i> (<code>com.github.librepdf:1.3.30</code>) for hand-built
JDBC connection pool	HikariCP	3.4.5	<code>com.zaxxer:HikariCP:3.4.5</code>	Fast pooled DB connect HikariCP 4.x/5.x need Java pin 3.4.5. (DAO layer, T
Password hashing (NFR-04, BR-02)	jBCrypt	0.4	<code>org.mindrot:jbcrypt:0.4</code>	Salted BCrypt hashing <code>PasswordHasher</code> .
MySQL driver	Connector/J	8.0.x	<code>mysql:mysql-connector-java:8.0.33</code>	JDBC access to MySQL fine on Java 8).
Logging	SLF4J + Logback	1.2.x	<code>ch.qos.logback:logback-classic:1.2.13</code>	Logback 1.2.x is the Java (1.3+/1.4+ need newer
Grid/complex forms (optional)	MigLayout for JavaFX (MigPane)	5.3	<code>com.miglayout:miglayout-javafx:5.3</code>	Only if <code>GridPane</code> becomes unwieldy on dense adn
UI testing (optional)	TestFX	4.0.16-alpha	<code>org.testfx:testfx-junit:4.0.16-alpha</code>	Automated screen test 4.0.16 line runs on Java

3.3 Libraries to AVOID on Java 8

These are commonly recommended but **require Java 11+** — do not add them while you stay on JDK 1.8:

- **FormsFX** and **ValidatorFX** (Java 11+) — use ControlsFX `ValidationSupport` + JFoenix `RequiredFieldValidator` instead.
- **Ikonli 11.x / 12.x** (Java 11+) — use FontAwesomeFX 8.15 or Ikonli **1.9.0**.
- **TilesFX** and recent **Medusa** gauges (Java 11+) — for the dashboard, build tiles from plain JavaFX `Region` / `VBox` cards, or use **Medusa 8.x** if you need gauges.
- **AtlantaFX**, **MaterialFX** (Java 11+/17+) — modern themes but not Java 8.

If you later adopt the TDD's Java 17 option, this whole table flips: you'd add `org.openjfx:javafx-*` 21, and could use FormsFX, ValidatorFX, Ikonli 12, AtlantaFX, TilesFX, HikariCP 5, etc. A one-page "upgrade delta" is in Appendix C.

4. Layout Architecture

4.1 How the UI maps onto MVC

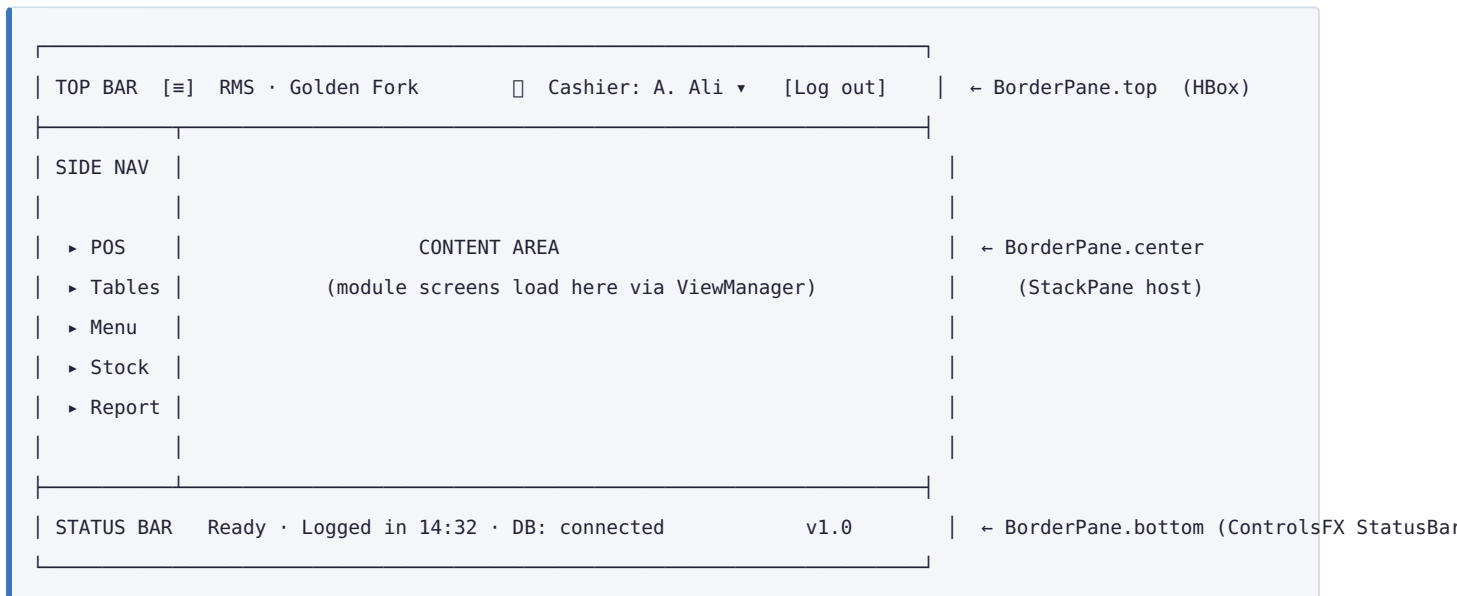
The parent TDD places JavaFX in the **View** and **Controller** tiers. Concretely:

FXML file (View)		Controller (thin)		Service (reused core)
LoginView.fxml	→	LoginController	→	AuthService.login()
PosOrderView.fxml	→	PosOrderController	→	OrderService / BillingService
MenuItemView.fxml	→	MenuItemController	→	MenuService
...		(event handling only)		(all rules, calc, SQL below)

Rule of thumb enforced throughout: **a controller may read fields, call one service method, and update nodes — nothing else**. If a controller is tempted to add, multiply, or validate a business rule, that logic belongs in a service.

4.2 The application shell

Every authenticated screen lives inside one persistent **shell** built on a `BorderPane`. Only the **center** region is swapped when the user navigates; the top bar and side navigation persist.



- **Top bar** — `HBox`: hamburger to collapse nav, app/restaurant name, notification bell (ControlsFX `NotificationPane`), current user + role menu, Log out (FR-04).
- **Side nav** — `VBox` of toggle buttons in a `ToggleGroup`. **Items are filtered by role at login** (see §5.2 and the FRD §2.4 matrix).
- **Content area** — a `StackPane` (`contentHost`) whose single child is replaced by the `ViewManager`.
- **Status bar** — ControlsFX `StatusBar` showing readiness, DB connectivity (NFR-08 relevance), and version.

`ShellController` owns the nav, the user context, and the `ViewManager`.

4.3 Screen loading / navigation strategy

A tiny `ViewManager` centralises FXML loading and content swapping. It is the *only* place `FXMLLoader` is used for module screens, which keeps navigation consistent and testable.

```
// util/ViewManager.java (View tier helper – no business logic)
public final class ViewManager {
    private final StackPane host;           // shell contentHost
    public ViewManager(StackPane host) { this.host = host; }

    public <C> C show(String fxmlPath) {    // e.g. "/fxml/pos/PosOrderView.fxml"
        FXMLLoader loader = new FXMLLoader(getClass().getResource(fxmlPath));
        Parent view = loader.load();        // throws -> caught, shown as error dialog
        host.getChildren().setAll(view);
        return loader.getController();      // caller may pass context (e.g. selected table)
    }
}
```

- **Login** is shown in its own `Scene` (no shell). On success, the shell scene replaces it.
- **Modals** (create/edit forms, confirmations) open as JFoenix `JFXDialog` over a `StackPane` overlay, or as a `Stage` with `Modality.APPLICATION_MODAL` for larger forms.

- **Background calls** (DB queries, report generation) run on a JavaFX **Task** so the UI never freezes; a ControlsFX **MaskerPane** covers the content while busy (supports NFR-02 perceived performance).

4.4 View package & resource structure

Extends the `com.rms.ui` package from TDD §2.3. Only this tree changes in Phase 2.

```
com.rms.ui
├─ ShellController.java
├─ ViewManager.java
├─ UiContext.java           // current Session/role handed to controllers (read-only)
├─ login/      LoginController
├─ pos/        PosOrderController, TableFloorController
├─ menu/       MenuCategoryController, MenuItemController
├─ inventory/  StockItemController, LowStockController
├─ purchasing/SupplierController, PurchaseOrderController, ReceiveDeliveryController
├─ staff/      StaffController
├─ reservation/ReservationController
├─ report/     SalesReportController, InventoryReportController, StaffActivityController
├─ admin/      UserAccountController, SystemConfigController
├─ common/     DialogFactory, Toast, FormValidators, TableCellFactories

src/main/resources
├─ fxml/       (mirrors the packages above: login/LoginView.fxml, pos/PosOrderView.fxml, ...)
├─ css/        rms.css (single theme) + optional pos.css for POS-only tweaks
├─ images/     logo.png, empty-states, print header
├─ i18n/       strings.properties (labels externalised for future localisation)
```

4.5 Layout container reference (which pane for which job)

Container	Used for	Example screen
<code>BorderPane</code>	Top/side/center/bottom frames	Shell; every module page root
<code>GridPane</code>	Labelled forms, aligned label:field rows	All create/edit forms
<code>VBox</code> / <code>HBox</code>	Vertical/horizontal stacks, toolbars, button rows	Side nav, action bars, order summary
<code>StackPane</code>	Overlays, dialogs, busy mask, content host	Content area, modal overlay
<code>TableView</code>	Tabular data with columns	Stock, suppliers, users, reports
<code>TabPane</code>	Grouping related sub-screens	Menu (Categories / Items), Reports
<code>TilePane</code> / <code>FlowPane</code>	Wrapping grids of equal cards/buttons	POS menu-item grid, table floor grid
<code>ScrollPane</code>	Content taller than the viewport	Long forms, order line list
<code>SplitPane</code>	Two resizable regions	POS (menu grid ↔ order ticket)

4.6 Resizing & responsiveness

- Minimum window **1024 × 768**; designed for **1366 × 768** (typical POS terminal). Layouts use panes + `HGrow/VGrow` and `maxWidth=Infinity` so they reflow rather than clip.
- POS uses a `SplitPane` so staff can widen the menu grid or the ticket.
- Data tables set sensible column resize policies (`CONSTRAINED_RESIZE_POLICY`) so columns fill width.
- Font sizes and hit targets are set in CSS (POS buttons ≥ 64 px tall) for touch terminals (NFR-01).

4.7 CSS theming strategy

- **One theme file** `rms.css` attached at the `Scene` level; every screen inherits it. Avoid inline `setStyle`.
- Define semantic style classes, not colours-in-place: `.rms-primary-btn`, `.rms-danger-btn`, `.status-free`, `.status-occupied`, `.low-stock-row`, `.total-amount`.
- **Status colours** map to enumerations (FRD Appendix B) and are always paired with a text label/icon (accessibility):

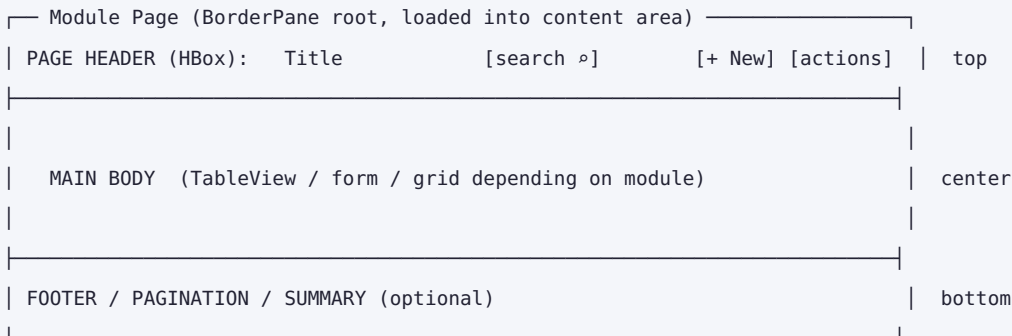
State	Class	Colour	Also shows
Table Free	<code>.status-free</code>	Green	"Free" + ○
Table Occupied	<code>.status-occupied</code>	Red	"Occupied" + ●
Table Reserved	<code>.status-reserved</code>	Amber	"Reserved" + ①
Table Needs Cleaning	<code>.status-cleaning</code>	Grey-blue	"Cleaning" + ☒
Low stock row	<code>.low-stock-row</code>	Amber background	⚠ + "Low"
Order Open / Paid / Cancelled	<code>.order-open/.order-paid/.order-void</code>	Blue / Green / Grey	status text

5. Global UI Components & Patterns

Reusable pieces live in `com.rms.ui.common` and are used by every screen so the app feels consistent.

5.1 Standard page template

Every module screen (except Login) uses the same skeleton so users always know where things are:



5.2 Role-filtered navigation (FR-02, FRD §2.4)

At login, `ShellController` receives the `Session` role and builds the nav accordingly. The action buttons inside each page follow the same rule.

Nav item	Administrator	Manager	Cashier
POS / Orders	✓	✓	✓
Tables (view/update)	✓	✓	✓
Reservations	✓	✓	✓
Menu & Prices	✓	✓	—
Inventory / Suppliers / Purchasing	✓	✓	—
Staff records	✓	✓	—
Reports	✓	✓	—
User Accounts & System Config	✓	—	—

A hidden item is also re-checked in the service layer (`RbacGuard.require`) — the UI hiding is convenience, not the security boundary (FRD FR-02, defence in depth).

5.3 Common dialogs & feedback (`DialogFactory`, `Toast`)

- **Confirmation** — JFoenix `JFXAlert` / `JFXDialog`: "Delete supplier 'X'? This cannot be undone." with Cancel / Confirm; destructive button uses `.rms-danger-btn`.
- **Form modal** — `Stage` (APPLICATION_MODAL) hosting a `GridPane` form (create/edit records).
- **Error** — typed service exceptions (`ValidationException`, `AuthorizationException`, `ConflictException`, `PersistenceException` — TDD §7.2) are caught by the controller and shown with the FRD's standardised messages.
- **Toast / notification** — ControlsFX `Notifications` for non-blocking success ("Order #1042 paid — receipt ready").
- **Busy overlay** — `MaskerPane` during background `Task`s.

5.4 Form & validation pattern (FRD Appendix A)

- Forms are `GridPane`: column 0 = right-aligned `Label`, column 1 = input control (`JFXTextField`, `ComboBox`, `Spinner`, `DatePicker`).

- **Inline validation:** JFoenix validators (`RequiredFieldValidator` , `NumberValidator`) + `ControlsFX ValidationSupport` . Errors show under the field in red; the Save button is disabled until valid.
- Field rules come **directly from FRD Appendix A** (username 3–50, price ≥ 0 two decimals, quantity integer ≥ 1 , discount % 0–100, reservation date must be future, etc.). The controller does presentation-level checks only; the service re-validates authoritatively.

5.5 Table (TableView) pattern

- Column set matches the data dictionary (TDD §3.4).
- Right-click / action column for Edit / Deactivate.
- Row styling via `rowFactory` — e.g. low-stock rows get `.low-stock-row` , inactive records greyed.
- Header filter/search box (ControlsFX) for long lists (stock, users).
- Empty state: a centered message + primary "New" button when the table is empty.

5.6 Visual language

- **Typography** — one sans-serif family (e.g. system default / "Segoe UI"/"Roboto"): 20 pt page titles, 13 pt body, 12 pt table cells; POS totals 28 pt bold.
- **Palette** — deep navy primary (matches your existing docs' headers), white surfaces, semantic status colours from §4.7, amber for warnings.
- **Icons** — FontAwesomeFX in nav and buttons (e.g. `fa-cash-register` POS, `fa-table` tables, `fa-boxes` stock), 18–20 px.
- **Spacing** — 8 px base unit; 16 px page padding; 12 px between form rows.

6. Screen-by-Screen Layout Design

Each screen below lists: the FXML root/containers, an ASCII wireframe, the key controls, RBAC, and the FRs it realises. Wireframes are indicative — proportions, not pixels.

6.1 Login (`login/LoginView.fxml`) — FR-01

Root: `StackPane` (background image/colour) → centered `VBox` card (JFoenix). Shown in its own scene, no shell.

The diagram shows a login form layout within a dashed border. Inside, there is a header section with a [LOGO] and the text 'RMS Login'. Below this are two input fields: 'Username' and 'Password', each followed by a placeholder line []. To the right of these fields are annotations: '← JFXTextField + RequiredFieldValidator' for the username field and '← JFXPasswordField (masked, BR-02)' for the password field. Below the password field is an error message '△ Invalid username or password' with the annotation '← generic error (FR-01 exception)'. At the bottom is a 'Sign in' button with the annotation '← JFXButton .rms-primary-btn'.

- **Controls:** `JFXTextField` username, `JFXPasswordField` password, `JFXButton` Sign in, inline error `Label`.
- **Behaviour:** Enter submits; button disabled until both fields non-empty; on the configurable failure count (default 5) the button is disabled with a "try again shortly" note (FR-01 throttle). Generic error message never reveals which field was wrong.
- **On success:** `AuthService.login()` returns a `Session`; the shell scene loads and routes to the role home (§6.2).
- **RBAC:** all roles. **Realises:** FR-01; enforces BR-01/BR-02 display rules.

6.2 Home dashboard (`ShellController` default content) — role-specific

Loaded into the content area right after login. Built from plain JavaFX cards (`VBox`/`TilePane`) — **no** `TilesFX` (Java 11+).

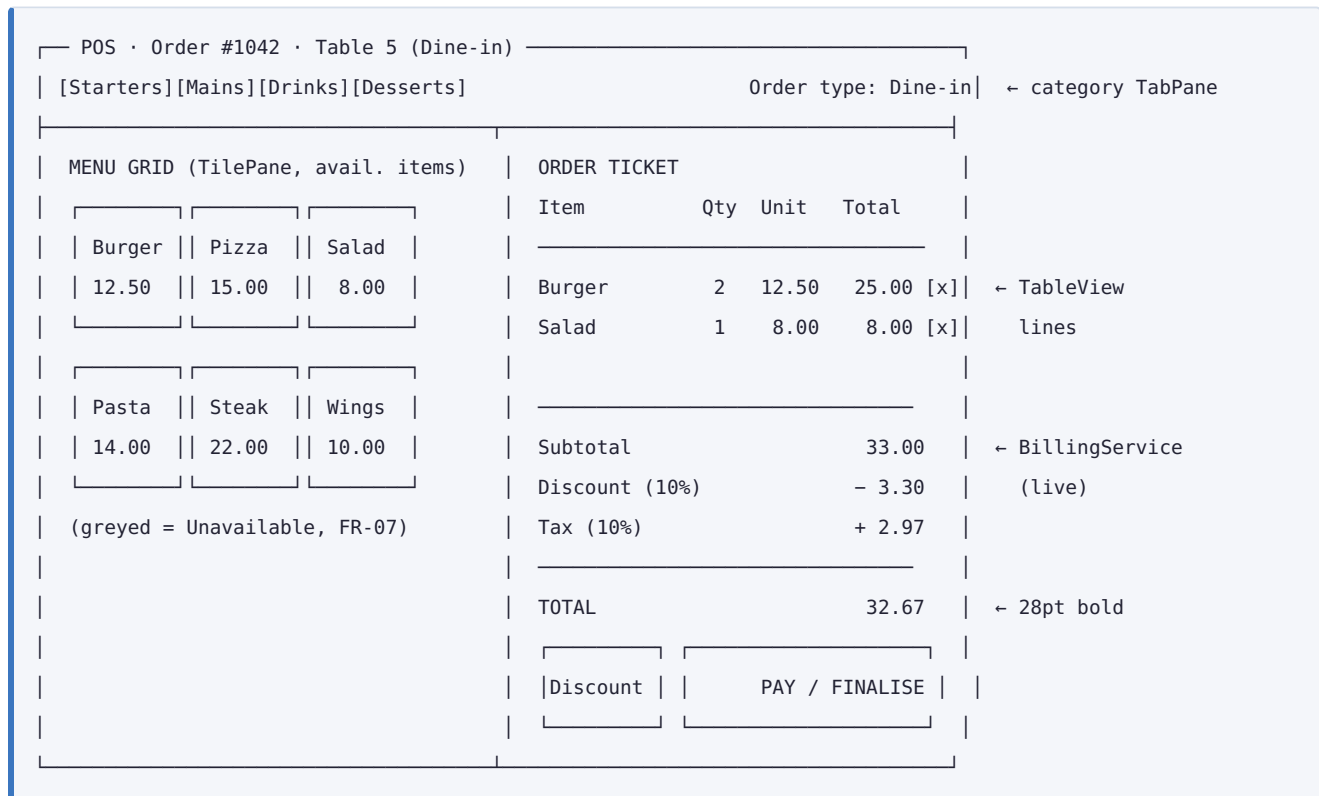
The diagram shows a home dashboard layout. At the top is a header bar with 'Home' on the left, 'Good afternoon, A. Ali · Cashier' in the center, and 'Fri 11 Jul' on the right. Below the header is a grid of four cards. The first card is 'Open orders' with the value '3'. The second card is 'Tables free' with the value '5 / 12'. The third card is 'Reservations' with the value 'today: 7'. The fourth card is '△ Low stock' with the value '2 items *'. Below the grid is a note '(* Manager/Admin)'. At the bottom is a 'Quick actions' section with three buttons: '[New order]', '[Table floor]', and '[New reservation]'.

- **Cards shown depend on role:** cashiers see orders/tables/reservations; managers/admins additionally see low-stock, today's sales, staff on shift.

- Each card is a clickable **VBox** navigating to its module. **Realises:** entry point to FR-09/10/22/24.

6.3 POS — Order entry & billing (`pos/PosOrderView.fxml`) — FR-10...FR-17 (the critical screen)

Root: `BorderPane`. **Center:** `SplitPane` — left = menu grid (`TilePane` of item buttons inside a `ScrollPane`, category tabs on top); right = the order ticket (`VBox` with a `TableView` of lines + totals panel). This is the highest-traffic screen, tuned for speed (NFR-01, BO-1).



- **Left (menu):** category `TabPane`; each tab a `TilePane` of `JFXButton` item cards (name + price). Tapping adds a line / increments qty (FR-11). Unavailable items are disabled + greyed (FR-07). Only `Available` items are tappable.
- **Right (ticket):** `TableView` of order lines (item, qty stepper, unit price snapshot, line total, remove ×). **Totals panel** below shows Subtotal / Discount / Tax / **Total**, all recomputed live by `BillingService` (never in the controller — FR-12/13/14, BR-13/16/17/18).
- **Discount** button opens a small modal: Percentage/Fixed toggle + value, validated 0–100 / ≤ subtotal (FR-13).
- **Pay / Finalise** opens the payment modal (§6.3.1).
- **Header** shows order number, table (or "Takeaway"), status. Opening the screen for a Free/Reserved table sets it Occupied (FR-10/09).
- **RBAC:** Cashier, Manager, Admin. **Realises:** FR-10, FR-11, FR-12, FR-13, FR-14 (display), FR-07 (availability).

Menu

[Categories | Items]

[+ New item]

Filter category: [Mains ▼]

Search: [pizza]

Name	Category	Price	Availability	Actions
Margherita	Mains	15.00	Available	[Edit] [Make unavail.]
Pepperoni	Mains	16.50	Available	[Edit] [Make unavail.]
Truffle Pizza	Mains	22.00	Unavailable	[Edit] [Make avail.]

← TableView

← greyed row

Item create/edit modal (GridPane form):

Edit menu item	
Name* [Margherita]	2-80, unique in category (FR-06)
Category* [Mains ▼]	existing category (FR-05)
Price* [15.00] (≥ 0, 2 dp)	FRD App.A validation
Availability [Available ▼]	FR-07
Description [...]	≤ 255
[Cancel] [Save]	

- Categories tab: simple **TableView** + add/edit/delete; delete of a non-empty category is blocked with guidance (FR-05).
- Price edits never touch finalised bills — a helper note reminds the manager (BR-09). Items with history are "Make unavailable", not deleted (BR-10).
- **RBAC:** Manager, Admin. **Realises:** FR-05, FR-06, FR-07.

6.6 Inventory / stock items (**inventory/StockItemView.fxml**) — FR-18, FR-22

Root: **BorderPane** → **TableView**; low-stock rows highlighted.

Inventory						[☒ Below reorder only] [+ New]
Name	Unit	On hand	Reorder	Status	Actions	
Tomatoes	kg	3.0 ▲	5.0	Active	[Edit] [Adjust] [Deact.]	← .low-stock-row
Mozzarella	kg	12.0	4.0	Active	[Edit] [Adjust] [Deact.]	
Olive oil	L	1.0 ▲	2.0	Active	[Edit] [Adjust] [Deact.]	← .low-stock-row

- On-hand is **read-only** here — it changes only via receipts (FR-21) or a logged **Adjust** action, never typed over (BR-21). The ▲ + amber row appears when on-hand ≤ reorder level (FR-22, BR-25).
- Create/edit modal: name (unique), unit of measure, reorder level (≥ 0), opening quantity (≥ 0).

- **RBAC:** Manager, Admin. **Realises:** FR-18, FR-22.

6.7 Suppliers (`purchasing/SupplierView.fxml`) — FR-19

Root: `BorderPane` → `TableView` + create/edit modal (name unique, contact/phone/email format-validated, status). Suppliers with PO history are set Inactive, not deleted (BR-22/BR-05). **RBAC:** Manager, Admin.

Suppliers					[+ New]	
Name	Contact	Phone	Email	Status	...	
Fresh Farms Co	M. Nabil	010-555-0134	sales@ff.co	Active	:	
Nile Beverages	S. Adel	010-555-0199	order@nileb.eg	Active	:	

6.8 Purchase orders (`purchasing/PurchaseOrderView.fxml` + `ReceiveDeliveryView.fxml`) — FR-20, FR-21

Two related screens in a `TabPane`: **Create PO** and **Receive delivery**.

Create PO — header (supplier, expected date) + a line editor `TableView` (add stock item + qty [+ unit cost]):

New Purchase Order				
Supplier* [Fresh Farms Co ▼]		Expected date [2026-07-15] (today+)		
└─ Lines (≥ 1) ─┐				
Stock item	Qty ordered	Unit cost	[+ add line]	
┌──────────┴──────────┴──────────┴──────────┐				
Tomatoes	20.0	1.20	[x]	
Olive oil	10.0	3.50	[x]	
└──────────┴──────────┴──────────┴──────────┐				
[Cancel] [Save PO (Ordered)]				

Receive delivery — pick an Ordered/Partially-received PO, enter received qty per line ($0 \leq \text{received} \leq \text{ordered}$). On confirm, `PurchasingService.receiveDelivery()` runs atomically: stock rises, PO status recomputes, low-stock flags refresh (FR-21, BR-24, NFR-03).

Receive delivery · PO-102 (Fresh Farms Co)				
Stock item	Ordered	Already recv	Receiving now	New on-hand
Tomatoes	20.0	0.0	[20.0]	→ 23.0
Olive oil	10.0	0.0	[6.0]	→ 7.0 (partial)
[Cancel] [Confirm receipt]				

- Creating a PO does **not** change stock (BR-23). Received > ordered is rejected (FR-21 exception).

- **RBAC:** Manager, Admin. **Realises:** FR-20, FR-21.

6.9 Staff records (`staff/StaffView.fxml`) — FR-23

Root: `BorderPane` → `TableView` + create/edit modal. Distinct from user accounts (BR-26); a staff row may or may not link to a login. Deactivate to keep history.

Staff					[+ New]	
Name	Position	Phone	Email	Status	Actions	
A. Ali	Cashier	010-555-0111	ali@gf.eg	Active	[Edit]	[-]
S. Mostafa	Waiter	010-555-0122	—	Active	[Edit]	[-]
H. Fouad	Chef	010-555-0133	hf@gf.eg	Inactive	[Edit]	← greyed

- **RBAC:** Manager, Admin. **Realises:** FR-23.

6.10 Reservations (`reservation/ReservationView.fxml`) — FR-24, FR-25, FR-26

Root: `BorderPane`. **Left/top:** a date picker + day's reservation `TableView` (or an agenda list).

Action: New reservation modal. Overlaps on the same table are blocked (FR-26/BR-27).

Reservations			Date [2026-07-11 ▼]		[+ New]		
Time	Customer	Party	Table	Status	Actions		
19:00	K. Mahmoud	4	T3	Booked	[Seat]	[Complete]	[Cancel]
19:30	L. Sami	2	T6	Booked	[Seat]	[Complete]	[Cancel]
20:00	N. Wael	6	T9	Seated	[Complete]	[Cancel]	

New reservation modal (GridPane):

New reservation		
Customer name*	[K. Mahmoud]	2-100
Contact*	[010-555-0150]	phone/email format
Date & time*	[2026-07-11] [19:00]	must be future (App.A)
Party size*	[4]	≥ 1; warns if > capacity (BR-28)
Table*	[T3 (cap 4) ▼]	existing table (FR-08)
[Cancel] [Save]		

| overlap on same table → rejected with conflict message (FR-26)

- Seating marks the table Occupied and can open an order; complete/cancel frees the reserved hold (FR-25/BR-29). Past date/time rejected; party > capacity warns/overrides (FR-24).
- **RBAC:** Cashier, Manager, Admin. **Realises:** FR-24, FR-25, FR-26.

6.11 Reports (report/*) — FR-27, FR-28, FR-29, FR-30

Root: `BorderPane` with a `TabPane`: **Sales, Inventory, Staff activity**. Each has a parameter bar (date range + filters), a results area (`TableView` + summary cards), and an **Export** button (PDF/print — FR-30). Manager/Admin only.

Reports

[Sales | Inventory | Staff activity]

From [2026-07-01] To [2026-07-11] Type [All ▼] Cashier [All ▼] [Run]

Summary

Sales 4,210.00 | Orders 118 | Avg 35.68 | Tax 382.7 | Discounts 96.5

Item / Category breakdown

Item	Qty	Revenue
Margherita	64	960.00
Burger	53	662.50

[+ Export PDF]

← params

← cards

← TableView

- Reports aggregate **finalised data only** (BR-30); figures come from stored order data, never current prices. `start ≤ end` validated; empty range shows a clear "no data" note (FR-27).
- **Inventory** tab: active stock with on-hand vs reorder + low-stock filter (FR-28). **Staff activity**: per-cashier logins, orders, sales, discounts (FR-29), RBAC-guarded.
- **Export** (FR-30) runs JasperReports on a background `Task`; the PDF header carries title, generated-by user, timestamp and the applied filters.
- **RBAC**: Manager, Admin. **Realises**: FR-27, FR-28, FR-29, FR-30.

6.12 User accounts (admin/UserAccountView.fxml) — FR-03 (Administrator only)

Root: `BorderPane` → `TableView` + create/edit modal (username unique 3-50, full name, role, initial password per policy, status). Cannot delete accounts with history (offer deactivate); cannot remove the last active Administrator (BR-06).

User accounts

[+ New user]

Username	Full name	Role	Status	Actions
admin	Owner	Administrator	Active	[Edit] [Deactivate]
m.hassan	M. Hassan	Manager	Active	[Edit] [Deactivate]
a.ali	A. Ali	Cashier	Active	[Edit] [Deactivate]

- **RBAC**: Administrator only (nav item hidden for others; service re-checks). **Realises**: FR-03.

6.13 System configuration (admin/SystemConfigView.fxml) — FR-14 admin data (Administrator only)

Simple `GridPane` form for reference/system data: **tax rate**, idle-timeout minutes, login max attempts. Values feed billing (tax rate snapshot) and session behaviour.

System configuration

Tax rate (%)	[10.00]	→ orders.tax_rate at finalise (BR-18)
Idle timeout (min)	[15]	→ auto-logout (FR-04)
Login max attempts	[5]	→ throttle (FR-01)
[Cancel] [Save]		

- **RBAC:** Administrator only. **Realises:** supports FR-14, FR-01, FR-04.

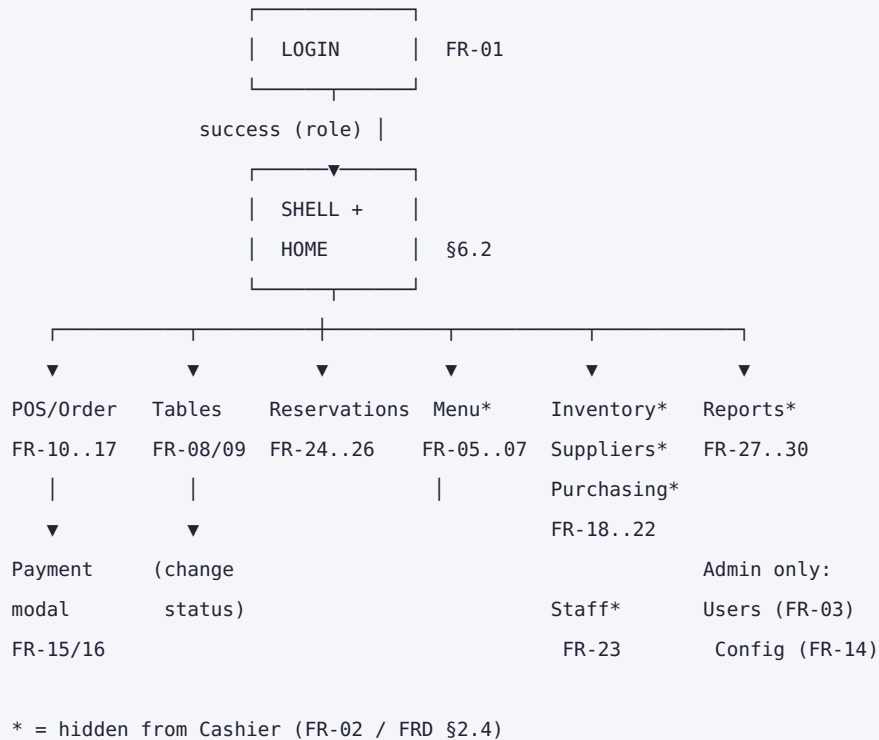
6.14 Receipt layout (report/ReceiptView → PDF) — FR-16

Not an interactive screen but a **rendered document** (JasperReports/OpenPDF), shown in a preview dialog with Print/Export. Reproduces the **stored finalised figures exactly** (BR-20); reprints are identical.

GOLDEN FORK			← restaurant header/logo
123 Nile St · Cairo			
Receipt / Order #1042			
2026-07-11 14:35			
Table 5 (Dine-in)			
Cashier: A. Ali			
2 Burger	12.50	25.00	← itemised lines (name, qty, unit, total)
1 Salad	8.00	8.00	
Subtotal		33.00	
Discount (10%)	-	3.30	
Tax (10%)	+	2.97	
TOTAL		32.67	
Cash		40.00	
Change		7.33	
Thank you!			

- **Realises:** FR-16 (and the export path shared with FR-30).

7. Navigation Map



- Nav persists; only the content area changes (§4.2). Deep links: Table floor → Order screen; Reservation "Seat" → Order screen; Inventory low-stock → Create PO.

8. Cross-Cutting UI Concerns

8.1 Keeping the View thin (NFR-05/06 — the migration guarantee)

- Controllers contain **only**: field reads, one service call, node updates, and navigation. Any `if` that encodes a business rule, any arithmetic, any SQL is a defect.
- Totals, tax, discounts, overlaps, stock math, RBAC decisions are **asked of services**, not computed in FXML controllers. This is what lets Phase 2 replace `com.rms.ui` wholesale (TDD §7.4) while `domain/service/dao` are reused verbatim.

8.2 Responsiveness & performance (NFR-02)

- All DB/report work runs on a JavaFX `Task`; the FX thread never blocks. Busy states use `MaskerPane`.
- `TableView` uses virtualised rendering (built-in) so large lists stay smooth.
- Perceived sub-2-second common actions: optimistic UI (add line immediately, reconcile on service return).

8.3 Accessibility & usability (NFR-01)

- Status never by colour alone — always icon + text.
- Full keyboard path for POS (number keys for qty, Enter to add, F-keys for pay).
- Consistent placement: primary action bottom-right, destructive actions confirm, labels above/beside every field.
- Labels externalised to `i18n/strings.properties` for future localisation.

8.4 Error & empty states

- Typed service exceptions → FRD standardised messages (e.g. "Username already exists", "Discount cannot exceed subtotal").
 - Every list has an empty state with a primary "New" call to action.
 - DB-unreachable is surfaced in the status bar and blocks finalise with a clear message (NFR-08/NFR-03).
-

9. Desktop → Web Migration Note (Phase 2)

Because this layout keeps the View strictly presentational, Phase 2 is additive (TDD §7.4):

- `domain`, `service`, `dao`, `util`, the MySQL schema, and the billing/reservation/stock logic are **untouched**.
 - Only `com.rms.ui` (FXML + controllers + `ViewManager`) is replaced by a web front end (HTML/JS) that calls the same services via a thin web-controller/REST tier.
 - The screen inventory, navigation map, RBAC matrix and status colour language in this document **carry over directly** as the spec for the web screens — so this document also seeds the Phase 2 UI design.
-

10. Appendices

Appendix A — JavaFX container cheat-sheet (what to reach for)

Need	Container / control
App frame with regions	<code>BorderPane</code>
Aligned label:field form	<code>GridPane</code>
Toolbar / button row	<code>HBox</code> (+ <code>Region</code> spacer with <code>HGrow=ALWAYS</code>)
Vertical stack / summary panel	<code>VBox</code>
Overlay / dialog / busy mask	<code>StackPane</code>
Grid of equal buttons/cards	<code>TilePane</code> / <code>FlowPane</code>
Two resizable panels	<code>SplitPane</code> (POS)
Tabular data	<code>TableView</code> + <code>TableColumn</code>
Grouped sub-screens	<code>TabPane</code>
Scrollable long content	<code>ScrollPane</code>
Numeric stepper	<code>Spinner</code>
Date entry	<code>DatePicker</code>
Material inputs/buttons	JFoenix <code>JFXTextField</code> / <code>JFXButton</code> / <code>JFXComboBox</code>
Toast / busy / validation	ControlsFX <code>Notifications</code> / <code>MaskerPane</code> / <code>ValidationSupport</code>

Appendix B — Consolidated Maven dependencies (JDK 1.8)

```

<!-- JavaFX 8 is bundled in the Java 8 JRE – no javafx dependency needed. -->
<dependencies>
  <dependency><groupId>com.jfoenix</groupId><artifactId>jfoenix</artifactId><version>8.0.10</version></dependency>
  <dependency><groupId>org.controlsfx</groupId><artifactId>controlsfx</artifactId><version>8.40.18</version></dependency>
  <dependency><groupId>de.jensd</groupId><artifactId>fontawesomefx</artifactId><version>8.15</version></dependency>
  <dependency><groupId>net.sf.jasperreports</groupId><artifactId>jasperreports</artifactId><version>6.20.6</version></dependency>
  <!-- or, lighter: <dependency><groupId>com.github.librepdf</groupId><artifactId>openpdf</artifactId><version>1.3.30</version></dependency>
  <dependency><groupId>com.zaxxer</groupId><artifactId>HikariCP</artifactId><version>3.4.5</version></dependency>
  <dependency><groupId>org.mindrot</groupId><artifactId>jbcrypt</artifactId><version>0.4</version></dependency>
  <dependency><groupId>mysql</groupId><artifactId>mysql-connector-java</artifactId><version>8.0.33</version></dependency>
  <dependency><groupId>ch.qos.logback</groupId><artifactId>logback-classic</artifactId><version>1.2.13</version></dependency>
  <!-- optional -->
  <dependency><groupId>com.miglayout</groupId><artifactId>miglayout-javafx</artifactId><version>5.3</version></dependency>
  <dependency><groupId>org.testfx</groupId><artifactId>testfx-junit</artifactId><version>4.0.16-alpha</version><scope>test</scope></dependency>
</dependencies>

<!-- Compile with maven.compiler.source/target = 1.8 -->

```

Appendix C — Upgrade delta if you move to Java 17 (from the TDD's stack)

Concern	Java 8 (this doc)	Java 17 / JavaFX 21 alternative
JavaFX	Bundled in JRE	<code>org.openjfx:javafx-controls/fxml:21</code> (+ modules)
Material controls	JFoenix 8.0.10	AtlantaFX / MaterialFX / MaterialFX-based themes
Icons	FontAwesomeFX 8.15 / Ikonli 1.9.0	Ikonli 12.x
Forms/validation	ControlsFX + JFoenix validators	FormsFX + ValidatorFX
Dashboard tiles/ gauges	Plain JavaFX cards / Medusa 8	TilesFX / Medusa (latest)
Connection pool	HikariCP 3.4.5	HikariCP 5.x
Scene Builder	8.5.0	Latest (17+)

Switching is confined to the presentation layer and build file — the reusable core is unaffected, consistent with NFR-06.

Appendix D — CSS style-class naming convention

`.rms-<component>-<variant>` for widgets (`.rms-primary-btn`, `.rms-danger-btn`, `.rms-card`), `.status-<state>` for status colours (matching FRD Appendix B enums), `.<context>-row` for row styling (`.low-stock-row`, `.inactive-row`). One `rms.css` ; no inline styles.

— End of Layout & UI Design TDD (RMS) v1.0 —